MILITARY SPECIFICATION

CHAFF ROLL (TYPE RR-( )/AL), GENERAL SPECIFICATION FOR

This specification is approved for use by the Naval Air Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification establishes the design, performance, and acceptance requirements for the Chaff Roll (Type RR-( )/AL). The chaff (radar reflector) roll or roll set contains components capable of creating confusing returns to radar systems operating within specified frequency ranges.

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

Federal

PPP-T-45	Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing
UU-P-268	Paper, Kraft, Wrapping
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-636	Box, Shipping, Fiberboard
QQ-S-781	Strapping, Steel, and Seals

Military

MIL-P-116	Preservation-Packaging, Methods of
MIL-B-131	Barrier Materials, Water Vaporproof, Flexible, Heat-Sealable
MIL-I-631	Insulation, Electrical, Synthetic-Resin Composition, Nonrigid
DoD-D-1000	Drawing, Engineering and Associated List
MIL-T-18303	Test Procedures; Preproduction, Acceptance and Life for Aircraft Electronic Equipment, Format for
MIL-C-85001/1	Chaff Roll Set (Type RR-171/AL), Specification Sheet
MIL-C-85001/2	Chaff Roll (Type RR-172A/AL), Specification Sheet

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
DoD-STD-480	Configuration Control-Engineering Changes, Deviations and Waivers

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MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-2071	Testing of Chaff Radar Cross-section

2.1.2 Other Government documents, drawing and publications. The following other Government documents, drawings, and other publications form a part of this specification to the extent specified herein.

DRAWINGS

Naval Air Systems Command

DL 1050AS158	Chaff Roll Assembly (Type RR-( )/AL)
DL 1050AS111	Chaff Roll, RR-( )/AL; Marking, Preservation, and Packaging of
DL 1050AS124	Vibration Fixture, Chaff Roll

MANUALS

Naval Air Systems Command

NAVAIR 16-30ALE41-1	Technical Manual, Maintenance Instructions, Intermediate, Dispensing Set, Countermeasures Chaff and AN/ALE-41

PUBLICATIONS

Bureau of Naval Weapons

WR-43	Preparation of Quality Assurance Provisions

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

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STANDARDS

American Society of Mechanical Engineers

ANSI Y14.5 Dimensioning and Tolerancing

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, N.Y. 10017)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS.

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheets, the latter shall govern.

3.2 First article. When specified in the contract, a sample shall be subjected to first article inspection (see 4.2, 6.2.1, and 6.3).

3.3 Chaff material. The chaff material shall be in accordance with the applicable specification sheet.

3.3.1 Dipoles. Dipoles shall meet the dimension and composition requirements of the applicable specification sheets. Dipoles composed of coated filaments shall have dc resistivity of less than 10 ohms per linear inch.

3.3.2 Coating. When required by the applicable specification sheet, a slip coating shall be applied to the chaff material to:

a. prevent dipole end welding and metal smear during the cutting operation

b. prevent cohesion between dipoles which may be caused by corrosive conditions.

c. increase dispersion by reducing dipole-to-dipole friction.

The formulation of the slip coating shall be approved by the procuring activity prior to use as part of the first article inspection procedures or quality conformance inspection procedures if first article inspection is not required (see 6.2.2).

3.3.3 Film. Chaff roll film shall be synthetic resin composition, non-rigid in accordance with MIL-I-631, Type G (Polyethylene terephthalate), Form F, Class II, clear translucent .002 inch thick by 12.500 ±.032 inches (unspliced).

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3.4 Design and construction. Each chaff roll (see 6.4.2), including chaff and all parts, assemblies, packing, and packaging; shall conform to all the requirements of this specification and applicable specification sheet for design, construction, and workmanship, except as otherwise specified in the contract (see 6.2.1). Configuration control shall be maintained in accordance with DoD-STD-480.

3.4.1 Nomenclature and identification marking. Nomenclature and identification marking shall be in accordance with MIL-STD-129, MIL-STD-130, and as specified herein.

3.4.2 Item definition. The chaff roll or roll set (see 6.4.3) covered by this specification shall consist of a standard roll or roll set filled with chaff components as specified in the applicable specification sheet.

3.4.3 Physical characteristics. The chaff roll or roll set shall be functionally compatible with the AN/ALE-38 and AN/ALE-41 (see 6.4.4) Dispensing Sets. Neither the operational reliability of either dispensing sets nor the effectiveness of the chaff shall be degraded when the chaff roll(s) or roll set(s) are used with these dispensers.

3.5 Standard conditions. The following conditions shall be used to establish normal performance characteristics under standard conditions.

a.	Temperature	Room ambient (25° ± 10°C)
b.	Altitude	Normal Ground
c.	Vibration	None
d.	Humidity	Room ambient up to 90 percent relative humidity

3.6 Service conditions. The chaff roll or roll set shall operate as required under any of the environmental service conditions or combinations of these conditions specified in MIL-STD-810 as modified herein. Tolerances shall be in accordance with the general requirements of MIL-STD-810.

3.7 Performance requirements. Unless otherwise specified herein, values set forth to establish the specified performance apply to performance under both standard and extreme service conditions. When reduced performance under extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under standard conditions will be specified herein.

3.7.1 Performance characteristics. Performance characteristics shall be in accordance with MIL-STD-2071 and the applicable specification sheet, when the chaff roll or roll set is tested in accordance with the requirements specified therein.

3.8 Detail requirements. The chaff roll or roll set contents shall meet the design, construction, and physical requirements contained herein. Drawings shall be interpreted in accordance with DoD-D-1000 and ANSI Y-14.5.

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3.8.1 Chaff roll set. A chaff roll set shall consist of identifiable chaff rolls, each of which shall conform to the requirements contained herein and the specific requirements of the applicable specification sheet.

3.8.2 Chaff roll.

3.8.2.1 Function. Each individual chaff roll shall be of the roll film type compatible with the AN/ALE-38 and AN/ALE-41 dispensers. The roll shall be formed by placing chaff between two layers of film to form a film/chaff/film sandwich which is subsequently wound on a feed shaft assembly to form a chaff roll.

3.8.2.2 Form factor. The film and chaff payload (see 6.4.7) shall have a length of 100 ±5 feet, and shall provide 51 ±1 revolutions on the roll hub. The maximum radius of the roll shall be 6.32 inches, and the minimum radius of the roll shall be 5.63 inches. Out of round conditions shall be held to a minimum and shall in no case be greater than .5 inch variation between the maximum and minimum radius measurements on the roll. The maximum allowable misalignment of film edges shall be in accordance with DL 1050AS158. The minimum gross weight of the film, hub, and chaff combination shall be as specified in the applicable specification sheet.

3.8.2.2.1 Packing density. Packing density shall conform with the requirements of the applicable specification sheet.

3.8.2.3 Leader. A leader, consisting of an extension of the two films of the film/chaff/film sandwich, shall be provided at the outer circumference of the chaff roll to facilitate loading of the roll in the ALE-38/ALE-41 cartridge. The length of each film leader shall be 20 +2, -0 feet.

3.8.2.4 Flipping of film. If the manufacturing process of the chaff roll is such that chaff is spread over two film layers and wound onto the feed shaft assembly, it shall be necessary to "flip" the films at the completion of winding each roll. This is accomplished by unwinding the outside layer of film until it meets the remaining film layer end, and cutting this unwound layer so that it is even with the remaining film end. Upon completion of this procedure, the outer film becomes the inner film and the inner film becomes the outer film, and thus provides proper film orientation for dispensing.

3.8.2.4.1 Marking of film. Regardless of the process used in manufacturing the roll, a label containing the words "FILM FLIPPED" shall be applied to the end of the film as an indication that the film/chaff/film sandwich is in the proper configuration for dispensing. In addition, the outer film of each chaff roll shall be marked with a 3/16 ± 1/16 inch rod stripe approximately 3 inches from and parallel to the leading edge of the film. The stripe shall run from the outer circumference of the film leader for a length of at least 80 feet toward the inner circumference of the chaff roll.

3.8.2.5 Chaff roll components. A chaff roll shall consist of the components specified in DL 1050AS158 and the applicable specification sheet.

3.8.2.6 Dipole configuration. The lengths, counts, and configurations of chaff dipoles included within each chaff roll shall be as specified in the applicable specification sheet.

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3.8.2.7 Weight. The minimum gross weight of a chaff roll shall be as specified in the applicable specification sheet.

3.8.3 Marking of chaff roll. Each chaff roll shall be permanently marked in accordance with MIL-STD-130, DL 1050AS158, and the applicable specification sheet.

3.9 Workmanship. Workmanship shall be in accordance with this specification and applicable specification sheet.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. The Government will conduct all flight tests.

4.1.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First Article Inspections (see 4.2)

b. Initial Production Inspection (see 4.3)

c. Quality Conformance Inspections (see 4.4)

4.1.2 Classification of characteristics. The characteristics verified by the tests and examinations described herein are classified as major or minor in accordance with WR-43. Test and examinations that verify major characteristics are identified by the symbol (M). Test and examinations described which are not annotated with a classification code are classified minor.

4.1.3 Lot information. A lot shall consist of a group of items of a specific configuration; produced within the same time frame at one location; by the same production line and production technique; and submitted for acceptance, inspection, or use at one time.

4.2 First article inspection. First article inspection shall be conducted by the contractor on three chaff rolls representative of the production units to be supplied under the contract. These chaff rolls shall be inspected according to the approved first article inspection procedure (see 4.5). The Government inspector and the procuring activity shall be advised when inspections are to be conducted so that a representative may be designated to witness or supervise the inspections when so desired. Contractors not having adequate facilities to conduct all required inspections shall obtain the services of a commercial testing laboratory acceptable to the Government. No first article inspections shall be conducted prior to approval of the first article test procedures by the contracting order. First article samples

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shall not be submitted for functional (Government) tests until in-house tests have been conducted in accordance with the approved test procedures, and the test results have been approved by the contracting officer.

4.2.1 First article inspection data. The contractor shall prepare and have available for review and approval a first article inspection report in accordance with MIL-STD-831 and the contract which includes all data collected during first article inspections (see 6.2.2).

4.2.2 Scope of inspections. At least three chaff rolls or roll sets shall be selected for first article inspection to verify that the chaff rolls meet all the requirements of this specification, the specification sheet, and the contract (see table I). In addition, one dispenser load (six chaff rolls) shall be selected for first article dispenser compatability tests as specified in the specification sheets. The tests on each sample set shall be performed in the order shown in table I. The group number shall be clearly marked on each roll (disk) before first article tests are started.

TABLE I. First article inspections - schedule and sequence.

Inspection	Requirement Paragraph	Inspection Paragraph	Chaff Roll Group		
			1	2	3
Chaff Materials	3.3	4.6	X		
Chaff Roll	3.8.2	4.7	X	X	X
High Temperature	3.6	4.8.1		X	
Low Temperature	3.6	4.8.2		X	
Temperature Shock	3.6	4.8.3		X	
Humidity	3.6	4.8.4		X	
Shock	3.6	4.8.5			X
Vibration	3.6	4.8.6			X
Functional	3.7	4.9	X	X	X

4.2.3 First article approval. Approval of the first article sample shall be by the procuring activity upon satisfactory completion of all inspections. No production chaff rolls shall be delivered prior to the approval of the first article sample.

4.2.4 Production chaff rolls. Chaff rolls (individual or sets) supplied under the contract shall in all respects, including design, construction, workmanship, performance, and quality be equal to the approved first article sample. Each chaff roll and assembly shall be capable of successfully passing the same tests as imposed on the first article sample. Evidence of noncompliance with the above shall constitute cause for rejection, and it

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shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial production inspections. One of the first 50 production rolls or roll sets shall be selected and sent at the contractor's expense to a designated Government laboratory for inspections. This sample shall be selected by the Government representative after it has successfully passed all individual inspections. No other inspections shall be conducted on the sample prior to starting the initial production inspections. The first article sample shall not be selected for this inspection.

4.3.1 Scope of inspections. The sample may be subjected to any and all inspections the procuring activity deems necessary to assure that the production equipment is equal to the previously approved sample in design, construction, workmanship, performance, and quality, and that it meets all requirements specified herein or in the contract.

4.3.2 Accessory material. In addition to the complete sample submitted for initial production inspections, the contractor shall also submit such accessory material and include data necessary to evaluate the sample in accordance with this specification and the contract.

4.3.3 Initial production sample approval. Approval of the initial production sample shall be by the procuring activity upon satisfactory completion of all inspections. Any design, material, or performance defect made evident during this inspection shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the initial production sample to pass any of the inspections shall be cause for deliveries under the contract to cease until proper corrective action is approved and accomplished.

4.4 Quality Conformance Inspections. The contractor shall furnish all samples and, with the exception of Government conducted flight tests, shall be responsible for accomplishing the quality conformance inspections. All examinations and tests shall be under the supervision of the Government representative. Contractors not having adequate facililties for conducting all required inspections shall engage the services of a commerical testing laboratory acceptable to the procuring activity. The contractor shall make available inspection reports showing quantitative results for all applicable quality conformance inspections (see 6.2.2). Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Quality conformance inspections shall consist of the following:

a. Individual Inspections (see 4.4.1)

b. Sampling Inspections (see 4.4.2)

c. Special Inspections (see 4.4.3)

4.4.1 Individual inspections. Individual inspections are those inspections conducted on each chaff roll. Individual inspections for the chaff rolls are specified in table II. Failure to pass any of these inspections shall be cause for rejection of the chaff roll.

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TABLE II. Individual inspections.

Inspection	Requirement Paragraph	Inspection Paragraph
Form Factor	3.8.2.2	4.7.1
Packing Density	3.8.2.2.1	4.7.2
Leader	3.8.2.3	4.7.3
Flipping of Film	3.8.2.4	4.7.4
Marking of Film	3.8.2.4.1	4.7.5
Chaff Roll Components	3.8.2.5	4.7.6
Dipole Layout	3.8.2.6	4.7.7
Marking of Chaff Roll	3.8.3	4.7.8
Packaging and Packing	5.3, 5.4	4.7.9

4.4.2 Sampling inspections. Chaff rolls selected for sampling inspections shall first have passed the individual inspections. Chaff rolls or roll sets shall be selected for sampling inspections by the Government representative in accordance with the schedule contained herein.

4.4.2.1 Scope of inspections. Sampling inspections shall include examinations and tests for verification of manufacturing and performance requirements.

4.4.2.1.1 Sampling inspection for verification of manufacturing processes. Sampling inspections for verification of the manufacturing processes shall include, as a minimum, those in-process inspections specified in 4.6 and 4.7.

4.4.2.1.2 Sampling inspections for verification of performance requirements. Sampling inspections for verification of performance requirements (Government conducted flight tests) shall be conducted on chaff rolls or roll sets to demonstrate conformance to the requirements of 4.9 and the applicable specification sheet when conducted in accordance with the procedures of MIL-STD-2071. Sampling shall be performed to provide two rolls or roll sets from each production lot for inspection.

4.4.3 Special inspections. Special inspections shall be conducted on a quantity of chaff rolls for the purpose of checking the effect of any design or material change on performance, and to assure adequate quality control. The chaff rolls selected for special inspections may be selected from those previously subjected to sampling inspections.

4.4.3.1 Special inspection schedule. Selection of chaff rolls or roll sets for special inspections shall be made as follows:

a. On early chaff rolls after an engineering or major change.

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b. Whenever failure reports or other information indicate that additional inspections are required. (This will be determined by the procuring activity.)

4.4.3.2 Scope of inspections. Special inspections shall consist of such examinations and tests as required by the procuring activity (see 6.2.2). Inspection procedures previously approved for the first article inspection shall be used where applicable. When not applicable, the contractor shall prepare an inspection procedure and make it available to the procuring activity prior to conducting the inspections (see 6.2.2).

4.4.4 Chaff roll failure. Should a failure occur during either the sampling or special inspections, the following action shall be taken:

a. Determine the cause of failure.

b. Determine if the failure is an isolated case or design defect.

c. Prepare and make available for procuring activity approval, the proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future inspections (see 6.2.2.).

d. Where practical, include an inspection in the individual inspections to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Inspection procedures. The procedures used for conducting first article inspection and quality conformance inspection shall be prepared by the contractor and made available to the procuring activity (see 6.2.2). The right is reserved by the procuring activity or the Government inspector to modify the inspections or require any additional inspections deemed necessary to determine compliance with the requirements of this specification and the contract (see 6.2.1). MIL-T-18303 shall be used as a guide for preparation of inspection procedures. When approved inspection procedures are available from previous contracts, such procedures may be used provided their use is acceptable to the procuring activity; however, the right is reserved by the procuring activity to require modification of such procedures, including additional inspections, when deemed necessary (see 6.2.1).

4.6 Inspection of materials.

4.6.1 Chaff material inspection. Examinations of the materials receiving inspection report for each receiving inspection lot (see 6.4.13), and on-line inspection of chaff-hank (see 6.4.9) and chaff-cut lots (see 6.4.10) as defined herein shall be made to adequately demonstrate compliance with the requirements of this specification and the applicable specification sheet.

4.6.1.1 Chaff-hank inspections.

4.6.1.1.1 Dipole characteristics inspection. Fifty fibers or strands from each chaff-hank lot (see 6.4.11) shall be inspected for the weight, dimension, and coating requirements of this specification and the applicable

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specification sheet. If more than 15 dipoles do not meet the specified requirements, the lot shall be rejected, and production stopped until corrective action has been taken. One chaff hank (see 6.4.9) from each chaff hank lot may be weighed, as a unit, in lieu of 50 fibers to verify the in-process weight requirement.

4.6.1.1.2 Secondary coating (M101). The formulation of the secondary coating shall be verified by means of appropriate chemical analysis. Two sample hanks (1 pound minimum) shall be randomly selected from each chaff-hank lot for this inspection. Samples failing to meet the approved coating formulation shall be cause for rejection of the chaff-hank lot.

4.6.1.2 Chaff-cut inspection.

4.6.1.2.1 Dipole length inspection (M102). Twenty dipoles from each chaff cut (see 6.4.8) shall be selected at random from each chaff-cut lot (specified herein). Each of these dipoles shall be measured, and its length recorded. If more than four dipoles from any cut, or more than 15 percent of the total sample are found to be out of the specified tolerance, that lot of chaff-cuts shall be rejected, and production stopped until corrective action has been taken. The inspection shall be repeated on the first chaff cuts processed after the corrective action has been taken to assure compliance with specification requirements.

4.6.1.3 Film. Examination of the film and receiving inspection report for the film shall be made to assure compliance with the specification requirements. Improper film composition or film dimensions shall be cause for rejection of the lot.

4.7 Chaff roll inspections.

4.7.1 Form factor.

4.7.1.1 Payload length (M103). Repeatable mechanical measurements shall be made to verify the payload length and the number of turns.

4.7.1.2 Roll radius (M104). Repeatable mechanical measurements shall be made to confirm compliance with the chaff roll radius requirements. Gages may be used for this measurement. In addition the first article roll radius shall be measured and recorded before and after vibration and shock exposures.

4.7.1.3 Out of roundness (M105). Out of roundness shall be measured to assure less than one-half inch variation in radius between minimum and maximum readings.

4.7.1.4 Film misalignment. The film misalignment shall be examined to verify that it is within tolerances specified in DL 1050AS158.

4.7.1.5 Chaff roll weight. The gross weight of film, hub assembly, and chaff combination shall be measured to verify the requirement of the applicable specification sheet.

4.7.2 Packing density. Chaff roll weight and dimensional examination shall confirm compliance with 3.8.2.2.1 herein.

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4.7.3 Leader. Verification of this characteristic shall be made by repeatable mechanical measurements.

4.7.4 Flipping of film (M106). This characteristic shall be verified during the manufacture of the roll. It shall also be verified by checking the position of the red strip after manufacture; i.e. the red stripe shall be in the outer mylar film after flipping.

4.7.5 Marking of film. The length of the red stripe shall be verified during the manufacture of the roll. An examination of the product shall be made to verify the presence of the "Film Flipped" label.

4.7.6 Chaff roll components. Verification of the presence of all components parts listed in DL 1050AS158 and the applicable specification sheet shall be made through visual inspection.

4.7.7 Dipole layout. Verification of compliance with the requirements of the applicable specification sheets and as contained herein shall be made during initial fabrication of the chaff roll immediately after the dipoles are placed on the film.

4.7.8 Marking of chaff roll. Visual examinations of the chaff roll shall be made to confirm compliance with 3.8.3.

4.7.9 Packaging and packing. Visual examination shall be made of the packaging, packing and markings to assure compliance with the requirements of Section 5 of this specification. Any discrepancies from the specification shall be cause for rejection.

4.8 Environmental tests. The simulated environmental tests specified below shall be performed in the order presented herein. Tolerances for these tests shall be in accordance with MIL-STD-810 unless otherwise specified herein. The chaff rolls and packaging supports, where applicable, shall not show signs of appreciable damage or degradation, and shall meet all dimensional and performance requirements after exposure.

4.8.1 High temperature. The chaff roll or roll set shall be subjected to a high temperature test in accordance with MIL-STD-810, Method 501.1, Procedure II, including Method 501.1 paragraph entitled, Summary; with the following exception: the maximum temperature shall be 85°C (185° Fahrenheit).

4.8.2 Low temperature. The chaff roll or roll set shall be subjected to a low temperature test in accordance with MIL-STD-810, Method 502.1, Procedure I, including Method 502.1 paragraph entitled, Summary; with the following exception: the minimum temperature shall be -65°C (-85° Fahrenheit).

4.8.3 Temperature shock. The chaff roll or roll set shall be subjected to a temperature shock test in accordance with MIL-STD-810, Method 503.1, Procedure I, including Method 503.1 paragraph entitled, Summary.

4.8.4 Humidity. The chaff roll or roll set shall be subjected to a humidity test in accordance with MIL-STD-810, Method 507.1, Procedure I, including Method 507.1 paragraph entitled, Summary.

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4.8.5 Shock. The chaff roll or roll set shall be subjected to a shock test in accordance with MIL-STD-810, Method 516.2, Procedure I, including Method 516.2 paragraph entitled, Summary.

4.8.5.1 Shock pulse. The pulse shape shall be as specified in figure 516.2-1 of MIL-STD-810 except that an electrodynamic vibration machine may be used, and if so, the tolerances of the specified pulse may be modified to allow amplitude variation up to ±20 percent of maximum amplitude prior to the initiation of the sawtooth ramp. All other tolerances apply. (The change in tolerance allows for the normal necessary pre- and post-pulses for electrodynamic machine shock excitation.)

4.8.5.2 Shock fixture. The shock fixture shall conform to DL 1050AS124. The control accelerometer location for each axis shall be as specified in DL 1050AS124. (Precautions should be taken to prevent loss of tension of the chaff roll film during loading and unloading. Should loss of tension occur, testing shall be invalid, and a new roll shall be tested).

4.8.6 Vibration. The chaff roll or roll set shall be subjected to a vibration test in accordance with MIL-STD-810, Method 514.2, Procedure IIA, including Method 514.2 paragraph entitled, Summary; except as follows:

a. The chaff roll must withstand both a Low Level Adequacy Test and a Critical Design Test.

b. Testing shall be performed in all three axis.

c. The same fixture used for the shock test shall be used for the vibration test.

d. The fixture shall be oriented in the flight-up position.

e. The control accelerometer location for each axis shall be as specified in DL 1050AS124.

4.8.6.1 Vibration test sequence.

a. Low Level Adequacy Vibration in accordance with figure 1 for 15 minutes in each axis.

b. Critical Design Vibration in accordance with figure 2 for 10 minutes in each axis.

c. Repeat step a.

4.9 Functional tests. All functional flight tests will be conducted by the Government in accordance with MIL-STD-2071, the applicable specification sheet, and as follows:

4.9.1 Flight tests. The Government performed flight tests will be conducted on first article, initial production, quality conformance, and special test units to verify performance capability. Tests will include Radar Cross-Section (RCS) test, in accordance with MIL-STD-2071 and as modified

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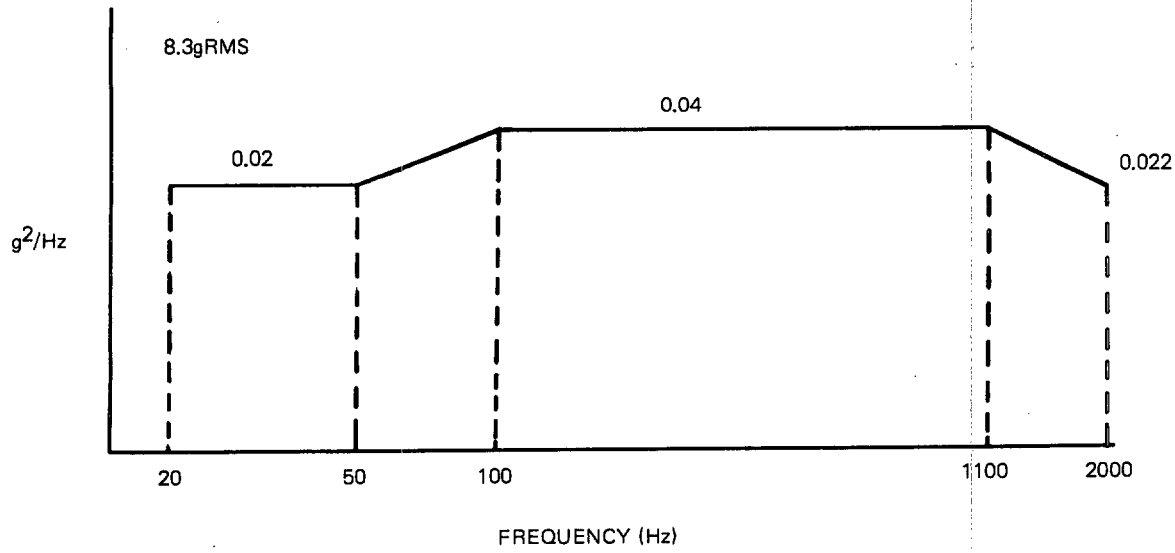


FIGURE 1. Low level adequacy vibration spectrum for chaff rolls.

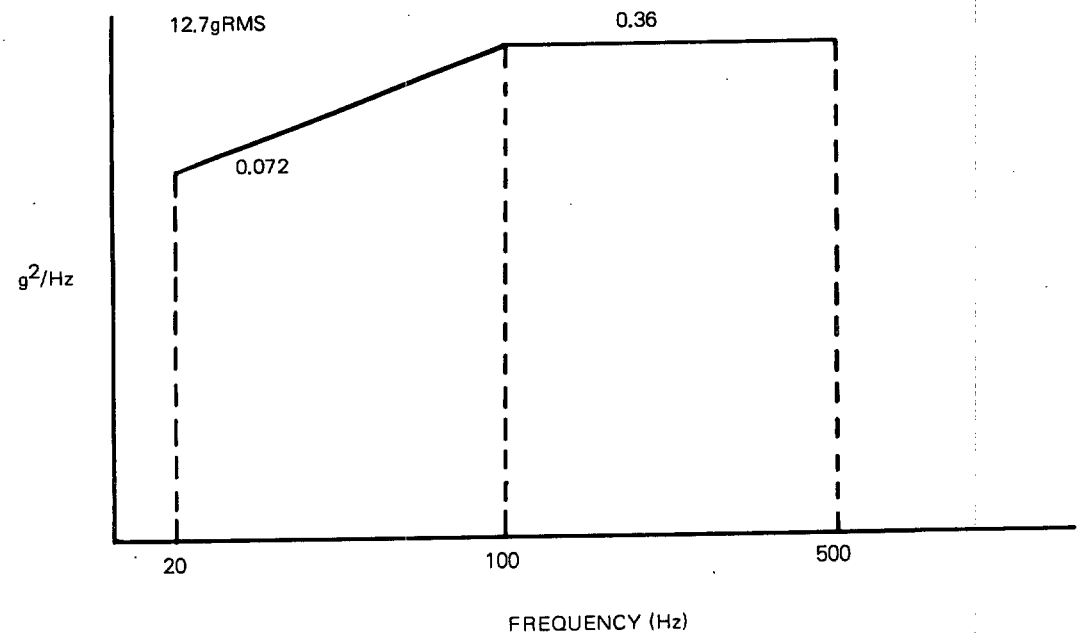


FIGURE 2. Critical design vibration spectrum for chaff rolls.

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herein. Chaff/dispenser compatibility tests will be conducted in addition to RCS tests for first article chaff rolls and chaff roll sets.

4.9.1.1 Aircraft. Flight tests will be conducted by the Government using a Navy A-7 aircraft. For test purposes, the aircraft will maintain an altitude of the number of feet and a true air speed of the number of knots specified in the applicable specification sheet (see 3.1).

4.9.1.2 Dispenser. The AN/ALE-41 dispenser should be used for all RR-( )/AL flight tests. If an ALE-41 is not available, an ALE-38 may be substituted; however, the ALE-38 will require modification for 115 volts ac power to operate on the Navy A-7 aircraft. All tests will be conducted at continuous dispense rate of 2 inches per second. The dispenser will be mounted on station number 1 of the aircrafts port wing.

4.9.1.3 Loading. The RR-( )/AL chaff rolls will be loaded into the AN/ALE-41 for dispensing in accordance with the procedures of NAVAIR 16-30ALE41-1 and the applicable specification sheet.

4.9.1.4 Radar. The RR-( )/AL chaff will be evaluated using a M-33 radar equipped in accordance with the applicable specification sheet (see 3.1).

4.9.1.5 Sampling. The test objective will be to conduct 10 runs of 40 seconds duration each; however, minimum test requirements can be met with 10 seconds of dispensing per run.

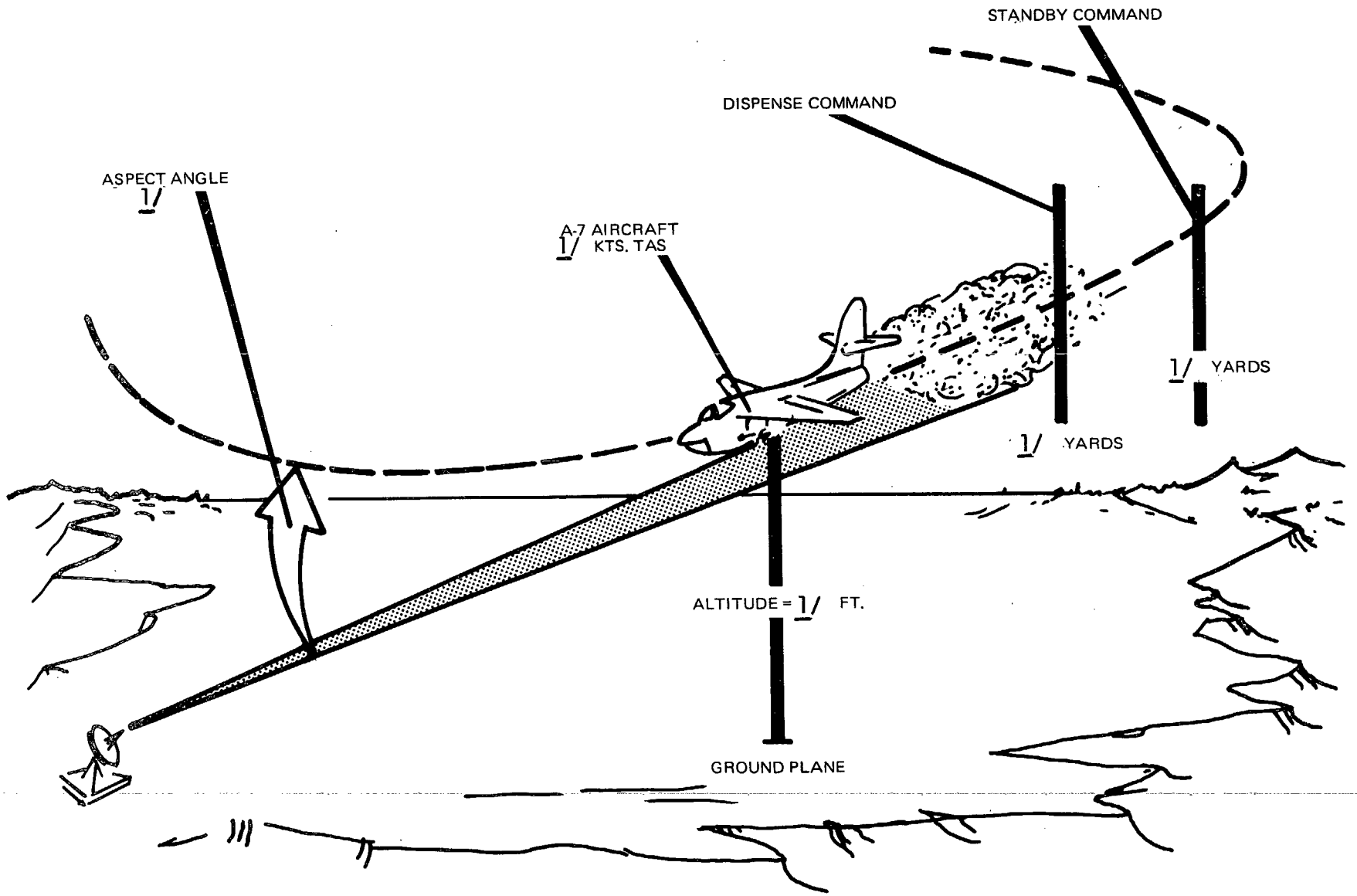
4.9.1.6 Special requirements, first article samples. First article samples at the completion of the test specified in 4.9.1.5 will be further tested for dispenser compatibility. Six rolls (independent or sets) will be loaded in a single ALE-41 dispenser for dispensing. The samples will then be tested at altitudes, airspeeds and dispense rates representative of operational usage to insure required performance of the complete chaff system.

4.9.2 Radar cross-section procedures.

4.9.2.1 Aircraft operation. The aircraft will fly a "racetrack" course as specified in figure 3. The aircraft will follow the station TACAN to fly a course directly over the test radar. Chaff will be dispensed on the inbound radial. Upon initial radar acquisition, the aircraft will be directed to maximum range, then brought in on a preliminary run. Since the leader on the chaff roll normally requires approximately 30 seconds of dispenser operation prior to chaff dispensing, the dispenser will be operated on the initial run to remove this leader.

4.9.2.2 Data gathering. The test radar will track the aircraft during the entire radial inbound run. Provisions will be made to prevent breaklock by installing a radar beacon or Luneberg lens on the aircraft. Areas of old chaff inadvertently crossed, or segments of runs in which the radar broke lock on the aircraft will be identified using the shadow graph, and rejected as invalid data. As depicted in figure 3, a standby and dispense command will be given at a range specified in the applicable specification sheet (see 3.1). Data recording will commence at the standby command and will continue for 10 seconds after termination of chaff dispensing. If old chaff creates an

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1/ Refer to applicable specification sheet for flight profile data.

FIGURE 3. Flight profile for testing of the RR-( )/AL Chaff Roll Set

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interference problem, a delayed start will be acceptable. The delayed start, however, may result in a short run.

4.9.2.3 Data evaluation. Data will be evaluated to determine the presence of at least 100 seconds of pulse-to-pulse data meeting the requirements of the procedures for collecting data on chaff immediately behind the aircraft, and the requirements for aspect angle as specified in MIL-STD-2071.

4.9.2.4 Data processing. Data will be processed to obtain RCS values of dispensed chaff in the radar resolution cells the number of feet behind the aircraft specified in the applicable specification sheet. Processing will be accomplished in accordance with MIL-STD-2071. The data will be normalized to 10° aspect angle and 0° antenna boresight gain.

4.9.2.4.1 Data selection. Shadow graphs will be used to select the data to be processed. For each run, the first record of data containing good RCS values in Gate 15 will be used as the start of the RCS data for that run. Normally, the next 20 test samples will be processed. Test samples of 1024 pulses will be used. The test samples to be processed for each data run will be contiguous unless the radar tracking errors necessitate breaking up the data samples in a run into two or three groups. In such cases, a total of 30 samples will be processed on each run if sufficient data are available. In all cases, data will be processed only when the radar is tracking properly as shown by the shadow graph results, and in no case will a hole or weak spot in the chaff cloud (caused by poor chaff performance) be eliminated from the data to be processed.

4.9.2.4.2 Calibration check. RCS measurement of a standard calibration sphere will be made prior to any chaff measurements. The sphere RCS data will be compared to the theoretical RCS of the sphere, and the calibration constant (ΔK) for the processor will be set to provide the measured sphere data as near as possible to the theoretical value of the sphere RCS. The corrected value of ΔK will be used in the first and record level data processing.

4.9.2.4.3 Correction factors. Correction factors will be in accordance with applicable specification sheet.

4.9.2.4.4 Statistical results. Results in the form of first level statistics will be accomplished by processing the selected data from the shadow graph using the same 1024 pulses to obtain the RCS values for all of the gates in each test sample. Shadow graphs will be used to define potential holes or weak spots in the chaff cloud, and when it is indicated that holes greater than those specified in 4.9.3.2 herein exist, the first level statistics of that portion of the data will be reprocessed in the following manner:

a. Test samples of 256 radar pulses will be used.

b. Reprocessing will start four test samples before the indicated hole in the first gate, and will continue two test samples beyond the hole as observed in the 15th Gate.

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c. Various sections along the test sample may be processed to define the exact severity of the holes or weak spots that are present.

4.9.3 Pass/Fail Criteria. The chaff rolls shall meet the requirements of 4.9.3.1 and 4.9.3.2 herein.

4.9.3.1 Radar cross-section (RCS). The average of the pulse-to-pulse RCS values as measured the number of feet specified in the applicable specification sheet behind the aircraft shall meet or exceed the number of square meters specified in the applicable specification sheet per gate for any unit to be acceptable. This average shall meet or exceed the number of square meters specified in the applicable specification sheet for at least 85 percent of the individual test runs.

4.9.3.2 Chaff cloud homogeneity. Discontinuities in the homogeneity of the chaff cloud will normally result in second level averages below those specified in 4.9.3.1. However, for acceptance purposes, discontinuities (holes) in the chaff cloud which exceed 0.5 seconds as displayed on the shadow graph, and which average less than the 50 percentile value of the first level processing of the run, will be cause for rejection of the chaff roll or roll set. The method for defining the width and depth of the hole will be as defined in 4.9.2.4.4 above.

4.10 Chaff dispenser compatibility test. The following procedures will be followed unless otherwise specified in the detailed test plan.

4.10.1 Flight profile. The flight profile will be specified in the detailed test plan. The altitude, airspeed, and dispense rate combinations used will be representative of operational usage.

4.10.2 Data gathering for chaff/dispenser compatibility test. The test radars will track the aircraft during the entire inbound run. Areas of chaff dispensed on previous runs that are inadvertently crossed, or segments of runs in which the test radar breaks lock on the aircraft, will be identified and assessed as to impact on the test results. If break-lock occurs, the radars will be oriented on the centroid of the chaff cloud, and the test procedures will continue from that point in time. When the aircraft is successfully tracked to the cessation of dispensing, the radars will remain on the terminal portion of the chaff cloud for 10 seconds, after which the radars will orient to the centroid of the chaff cloud for three additional minutes. RCS data recording will commence 10 seconds before chaff dispensing begins, and continue until 10 seconds after dispensing has ceased. Thereafter, 10 seconds recordings of RCS data will be taken for each of the three additional minutes that the chaff cloud is monitored. At each recording interval, the range and azimuth/elevation angles of the radar will be recorded. Immediately following the flight test, the chaff pod will be removed from the aircraft and inspected in the laboratory. Inspection results will be entered on the log sheet as specified in MIL-STD-2071.

4.10.3 Pass/fail criteria for chaff dispenser compatibility test. The chaff rolls shall meet the requirements of 4.10.3.1 and 4.10.3.2 herein. A numerical value for chaff cloud radar cross-section is not a part of the pass/fail criteria for this test, although this data should be maintained for data accumulation purposes.

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4.10.3.1 Roll jams. Jams, not attributed to loading errors or dispenser failures, in excess of one roll-jam per full pod (6 rolls) will be cause for rejection of the first article sample submission.

4.10.3.2 Chaff cloud discontinuities (chaff dispenser compatibility test). Discontinuities in the homogeneity of the chaff cloud which exceed those described in 4.9.3.2 herein will be cause for rejection of the first article sample submission.

5. PACKAGING

5.1 General. All major units and parts of the equipment shall be preserved, packaged, packed, and marked for the level of shipment specified in the contract or purchase order (see 6.2.1) in accordance with MIL-STD-794 and as specified herein. Unless otherwise specified in the contract or purchase order, the method of preservation-packaging shall be Method I, Submethod IA-15 in accordance with MIL-P-116 and this specification (see 6.2.1).

5.2 Packaging of roll in inner carton. One chaff roll shall be packaged per inner carton. The chaff roll shall be supported by the roll hub. Two plywood panels secured by a strap shall support the roll hub, and two additional blocks shall be used as spacers to protect the drive spline as specified in DL 1050AS111.

5.2.1 Carton details. The inner carton shall be a Type CF, Class "Weather-Resistant", Variety SW, Grade V3C container (RSC) conforming to PPP-B-636 and DL 1050AS111. The bottom longitudinal closure of the carton shall be completed by using a suitable length of filament reinforced tape, Type II conforming to PPP-T-45, with a minimum width of 3 inches. The four end joints formed by the flaps shall be sealed with 60 pound Kraft paper tape with a minimum width of 3 inches. The top closure shall be completed after the carton is filled by using a suitable length of filament reinforced tape, Type II conforming to PPP-T-45, with a minimum width of 3 inches. A 6 inch length of this tape shall be left free at one end of the tape and folded back and sealed to itself for a length of 3 inches, to provide a handle for grasping to use the tape as a ripcord in unsealing the top longitudinal flaps. The four end joints formed by the flaps shall be sealed with 60 pound Kraft paper tape with a minimum width of 3 inches. Cartons shall conform to the test requirement specified herein.

5.2.2 Barrier material. The carton containing the chaff roll shall be enclosed in a one-piece wrapping of water vapor proof, oil resistant barrier material, Class 1, conforming to MIL-B-131. The barrier material shall be sealed by the application of heat as specified in MIL-B-131.

5.3 Packaging details.

5.3.1 Interior packaging. All interior cartons shall be packaged in accordance with MIL-P-116 and DL 1050AS111.

5.3.2 Exterior packaging. Consignments shall be prepared for either domestic or overseas shipments as specified by the procuring activity.

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5.3.3 Marking. The inner carton and barrier material shall be permanently stamped in waterproof, petroleum resistant ink as specified in MIL-STD-129 and DL 1050AS111.

5.4 Packing (domestic and overseas packing of material in cartons). Twelve of the above cartons meeting the requirements of DL 1050AS111 shall be placed in a wooden box, style A, B, or I, constructed in accordance with PPP-B-601. A case liner composed of Kraft paper meeting the requirements of UU-P-268 shall be used.

5.4.1 Marking for shipments. Shipping containers shall be marked in accordance with 5.3.3 and as specified in the contract (see 6.2.1).

5.4.2 Palletizing. Palletizing shall be accomplished in accordance with DL 1050AS111. Exterior banding shall be type I or IV, heavy duty, Finish B (zinc-coated), Grade 2 as specified in QQ-S-781.

6. NOTES

6.1 Intended use. Chaff rolls or roll sets covered by this specification are intended for use in the AN/ALE-38 or AN/ALE-41 Dispensing Set.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification

b. Chaff roll type (see item description of applicable specification sheet)

c. Levels of preservation - packaging, packing and marking required (see section 5)

d. If design, construction or workmanship are to be other than specified in specification (see 3.4)

e. If first article is required (see 3.2, 6.3).

f. If test procedures are to be expanded or modified (see 4.5)

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below will be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

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Paragraph no.	Data requirement title	Applicable DID no.	Option
3.3.2, 4.5	Procedure, First Article Inspection	DI-T-4901	
4.2.1	Report, First Article Inspection	DI-T-4902	
4.4	Report, Production Inspection	DI-T-4904	
4.4.3.2, 4.5	Procedure, Production/Acceptance Inspection	DI-T-4903	
4.4.4c	Failure Data Collection, Analysis and Corrective Action Plan	UDI-T-23719	

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The first article sample should consist of three chaff rolls. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination and test approval of the document's first article.

6.4 Definitions. The following definitions apply to this specification:

6.4.1 Cartridge. A mechanical device which serves as the interface between a Chaff Roll and the ALE-38/ALE-41 dispensers.

6.4.2 Chaff roll. The chaff package configuration containing the chaff payload for use with the ALE-38/ALE-41 dispensers.

6.4.3 Chaff roll set. This term refers to two or more chaff rolls containing different configurations of dipole cuts, but manufactured as a set under one "RR" designation.

6.4.4 AN/ALE-38, AN/ALE-41. This nomenclature refers to one complete pod including nose cone, tail cone, case, six cartridges, and dispensing mechanism.

6.4.5 System. As used herein refers to one ALE-38/ALE-41 described above.

6.4.6 Pod. Pod is used to define the structural case of the ALE-38/ALE-41 and shall include the nose and tail cones.

6.4.7 Payload. The payload is the chaff carried in the chaff roll.

6.4.8 Chaff cut. A quantity of dipoles cut from a chaff hank to provide one row of dipole elements within a chaff roll.

6.4.9 Chaff hank. An assemblage of drawn metallized glass fibers produced by one machine prior to the dipole cutting operation. The number of

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fibers or strands in the chaff hank is normally equivalent to the number of dipoles required for each caff cut within a specific chaff roll.

6.4.10 Lot, chaff-cut. A chaff-cut lot shall consist of all chaff-cuts produced by one cutting machine for use in the chaff roll payload. The lot will contain as a maximum that quantity of chaff-cuts necessary to produce 100 chaff rolls.

6.4.11 Lot, chaff hank. A chaff-hank lot will consist of all chaff hanks produced by the same production machine and submitted for inspection at one time. This lot will not contain more hanks than are required to produce one lot of chaff rolls.

6.4.12 Lot, chaff roll. The chaff roll lot will contain the quantity of rolls or roll sets specified in the contract or purchase order.

6.4.13 Lot, receiving inspection. A receiving inspection lot will consist of all material of one type received from one supplier at one time.

6.5 Preparing the chaff roll for vibration and shock. The installation of the chaff roll into the fixture is designed to simulate the actual mounting in the aircraft pod. This mounting is fairly simple, but it does involve the use of a number of component parts. The following is included herein to facilitate and standardize the mounting method.

a. Equipment

(1) Chaff Roll Cartridge.

(2) Chaff Roll Fixture (with Mounting Shafts (2), Dogbone Cover Clamps (Feed Shaft Clamps) (2), Take-up Shaft Supports (2), and Take-up Shaft Splined Locks (2)).

(3) Male Spindle (Splined Dogbone), Part No. 1050AS121 of DL 1050AS158 (in addition to the one which normally comes with the Chaff Roll).

(4) Two Female Spindles (Dogbone Covers), Part No. 1050AS157 of DL 1050AS158.

(5) Miscellaneous Hex Head Bolts (and possibly spacers - see 7 below).

b. Procedure:

(1) Install the Chaff Roll in the Chaff Roll Cartridge and start the films on the Take-up Shafts in the normal manner.

(2) Install the additional Splined Dogbone in the Chaff Roll and place the Dogbone Covers over the exposed splines of the two Splined Dogbones.

(3) While holding the Mounting Shafts in the U-shaped channels (Intercostal Assemblies) of the Chaff Roll Loading Cartridge,

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place the Loading Cartridge and Chaff Roll into the Chaff Roll Fixture.

(4) Secure the Mounting Shafts and, after centering the Chaff Roll in the Fixture, secure the Dogbone Cover Clamps. The Dogbone Covers should be pressed against the Chaff Roll on each side to simulate in-service conditions.

(5) Place the Take-up Shaft Supports into the holes in the Fixture at the non-splined ends of the Take-up shafts, and secure.

(6) Position the Take-up Shaft Splined Locks over the Take-up Shaft splines and turn each Lock in the direction of Take-up Shaft rotation until the Take-up Shaft slip clutches have applied maximum torque to the Take-up Shafts.

(7) Secure the Take-up Shaft Splined Locks. It may be necessary to add spacers between the side of the fixture and the lip of the Take-up shaft Splined Locks, because the Take-up shafts should fit snugly between the Take-up Shaft Supports and Take-up Shaft Splined Locks without over-compression.

(8) Recheck Roll centering so as to insure a snug fit at each end of the chaff Roll center shaft and at both ends of each Take-up Shaft. Make adjustments as necessary, but be sure that the Take-up Shafts have applied maximum tension to the Chaff Roll films before testing begins. Recheck Chaff Roll film tension after each phase of vibration and shock.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-85001B(AS)	Chaff Roll (Type RR-( )/AL), General Specification for

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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